



Angle-Closure Glaucoma

(Closed-Angle Glaucoma)

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Angle-closure glaucoma is glaucoma associated with a physically obstructed anterior chamber angle, which may be chronic or, rarely, acute. Symptoms of acute angle closure are severe ocular pain and redness, decreased vision, colored halos around lights, headache, nausea, and vomiting. Intraocular pressure (IOP) is elevated. Immediate treatment of the acute condition with multiple topical and systemic medications is required to prevent permanent vision loss, followed by the definitive treatment, iridotomy.

(See also [Overview of Glaucoma](#).)

Angle-closure glaucoma accounts for approximately 10% of all glaucomas in the United States ([1](#)).

Reference

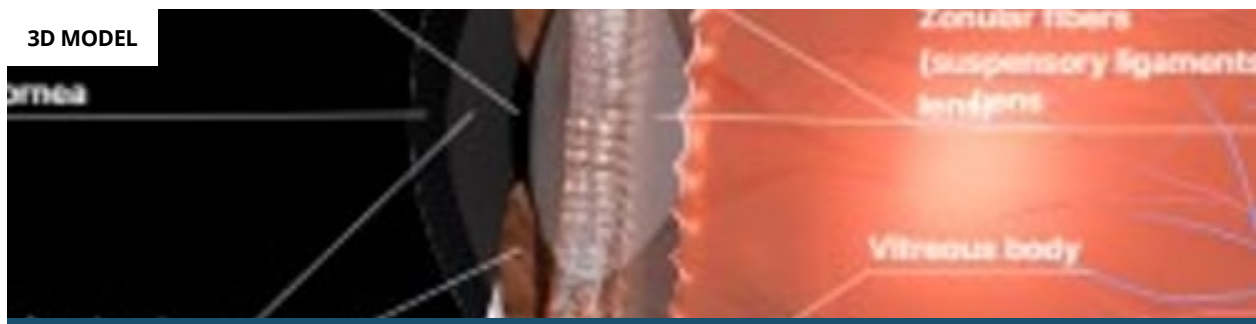
1. [Quigley HA, Vitale S](#). Models of open-angle glaucoma prevalence and incidence in the United States. *Invest Ophthalmol Vis Sci*. 1997;38(1):83-91.

Etiology of Angle-Closure Glaucoma

Angle-closure glaucoma is caused by factors that either pull or push the iris up into the angle (ie, junction of the iris and cornea at the periphery of the [anterior chamber](#)), physically blocking drainage of aqueous and raising intraocular pressure (IOP) (see table [Angle-Closure Glaucoma](#)). Elevated IOP damages the optic nerve.

Pathophysiology of Angle-Closure Glaucoma

Eye: Anterior and Posterior Chambers



Angle closure may be primary (cause is unknown) or secondary to another condition and can be acute, subacute (intermittent), or chronic.

Primary angle-closure glaucoma

Narrow angles are not present in young people. As people age, the lens of the eye continues to grow. In some but not all people, this growth pushes the iris forward, narrowing the angle. Risk factors for developing narrow angles include family history, advanced age, and race; risk is higher in Asian and Inuit populations and lower in European and African populations.

In people with narrow angles, the distance between the iris at the pupil and the lens is also very narrow. When the iris dilates, forces pull the iris centripetally and posteriorly, causing increasing iris–lens contact which prevents aqueous from passing between the lens and iris, through the pupil, and into the anterior chamber (this mechanism is termed pupillary block). Pressure from the continued secretion of aqueous into the posterior chamber by the ciliary body pushes the peripheral iris anteriorly (causing a forward-bowing iris called iris bombe), closing the angle. This closure blocks aqueous outflow, resulting in rapid (within hours) and severe (> 40 mm Hg) elevation of intraocular pressure (IOP). Nonpupillary block mechanisms include plateau iris syndrome, in which the central anterior chamber is deep but the peripheral anterior chamber is made shallow by a ciliary body that is displaced forward.

Acute angle-closure glaucoma requires immediate recognition and treatment because vision can be lost quickly and permanently.

Intermittent angle-closure glaucoma occurs if the episode of pupillary block resolves spontaneously after several hours, usually after sleeping supine.

Chronic angle-closure glaucoma occurs if the angle narrows slowly, allowing scarring between the peripheral iris and trabecular meshwork; IOP elevation is slow.

Pupillary dilation (mydriasis) can push the iris into the angle and precipitate acute angle-closure glaucoma in any person with narrow angles.

Secondary angle-closure glaucomas

In secondary angle-closure glaucomas, the mechanical obstruction of the angle is due to a coexisting condition, such as [proliferative diabetic retinopathy](#) (PDR), ischemic [central vein occlusion](#), [uveitis](#), or epithelial down-growth. Contraction of a neovascular membrane (eg, in PDR) or inflammatory scarring can pull the iris into the angle.

Symptoms and Signs of Angle-Closure Glaucoma

Acute angle-closure glaucoma

Patients have severe ocular pain and redness, decreased vision, colored halos around lights, headache, nausea, and sometimes vomiting. The systemic complaints may be so severe that patients are misdiagnosed as having a neurologic or gastrointestinal problem. Examination typically reveals conjunctival hyperemia, a hazy cornea, a fixed mid-dilated pupil, and anterior chamber inflammation. Vision is decreased. Intraocular pressure (IOP) is usually 40 to 80 mm Hg. The optic nerve is difficult to visualize because of corneal edema, and visual field testing is not done because of discomfort. For primary mechanisms of angle-closure (eg, pupillary block and plateau iris), examination of the

uninvolved contralateral eye can indicate the diagnosis.

Pearls & Pitfalls

- In patients who have sudden headache, nausea, and vomiting, examine the eyes.

Chronic angle-closure glaucoma

This type of glaucoma manifests similarly to [open-angle glaucoma](#). Some patients have ocular redness, discomfort, blurred vision, or headache that lessens with sleep (perhaps because of sleep-induced miosis and posterior displacement of the lens by gravity). On gonioscopy, the angle is narrow.

Peripheral anterior synechiae (PAS) (adhesions between the peripheral iris and angle structure) may be seen and cause blockage of trabecular meshwork and/or ciliary body face. IOP may be normal but is usually higher in the affected eye.

Diagnosis of Angle-Closure Glaucoma

- **Acute:** Measurement of intraocular pressure (IOP) and clinical findings
- **Chronic:** Gonioscopy showing peripheral anterior synechiae and characteristic optic nerve and visual field abnormalities

Diagnosis of acute angle-closure glaucoma is clinical and by measurement of IOP. Gonioscopy may be difficult to do in the involved eye because of a clouded cornea with friable corneal epithelium. However, examination of the other eye reveals a narrow or occludable angle. If the other eye has a wide angle, a diagnosis other than primary angle-closure glaucoma should be considered. Pupillary block mechanism can be recognized as iris bombe, an anterior bowing or arc of the iris between the angle and the pupil.

Diagnosis of chronic angle-closure glaucoma is based on the presence of peripheral anterior synechiae on gonioscopy and characteristic optic nerve and visual field changes (see symptoms and signs of [primary open-angle glaucoma](#)).

Before giving or prescribing mydriatic eye drops (which can trigger acute narrow-angle glaucoma in patients at risk), anterior chamber depth should be assessed. If gonioscopy is unavailable, the chamber depth can be more crudely assessed using a slit lamp or even a penlight; the penlight is held next to the temporal side of the eye, with the light beam parallel to the iris. If a shadow is cast on the nasal iris, the angle is probably narrow, because the temporal iris bows forward and blocks the light.

Treatment of Angle-Closure Glaucoma

- **Acute:** [Timolol](#), [pilocarpine](#), and [brimonidine](#) drops, oral [acetazolamide](#), and a systemic osmotic agent, followed by laser peripheral iridotomy
- **Chronic:** Similar to primary open-angle glaucoma except that laser peripheral iridotomy should be done if the ophthalmologist feels that the procedure may slow mechanical closing of the angle. [Cataract](#) removal helps delay the progression of chronic angle-closure glaucoma.

Acute angle-closure glaucoma

Treatment must be initiated immediately because vision can be lost quickly and permanently. The patient should receive several medications as soon as possible. A suggested regimen is [timolol](#) 0.5% one drop every 30 minutes for 2 doses, [pilocarpine](#) 2 to 4% one drop every 15 minutes for 2 doses, [brimonidine](#) 0.15 or 0.2% one drop every 15 minutes for 2 doses, [acetazolamide](#) 500 mg orally initially (IV if patients are nauseated) followed by 250 mg every 6 hours, and consider an osmotic agent, such as oral glycerol 1 mL/kg diluted with an equal amount of cold water, [mannitol](#) 1.0 to 1.5 mg/kg IV, or isosorbide 100 g orally (220 mL of a 45% solution). (**NOTE:** This form of isosorbide is not [isosorbide dinitrate](#).) Response is evaluated by measuring intraocular pressure (IOP). Miotics (eg, [pilocarpine](#)) are generally not effective when IOP is > 40 or 50 mm Hg because of an anoxic pupillary sphincter.

Definitive treatment is with laser peripheral iridotomy (LPI), which opens another pathway for fluid to pass from the posterior to the anterior chamber, breaking the pupillary block. It is done as soon as the cornea is clear and inflammation has subsided. In some cases the cornea clears within hours of lowering the IOP; in other cases, it can take 1 to 2 days. LPI is done on both eyes because the chance of having an acute attack in the other eye is 80%.

The risk of complications with LPI is extremely low compared with its benefits. Glare, which can be bothersome, may occur. If LPI can not be performed, a surgical iridectomy may be needed.

Chronic angle-closure glaucoma

Patients with chronic, subacute, or intermittent angle-closure glaucoma should also have LPI. Additionally, patients with a narrow angle, even in the absence of symptoms, should undergo prompt LPI to prevent angle-closure glaucoma. If a cataract is present, [cataract removal](#) can dramatically delay the progression of chronic angle-closure glaucoma.

The medications and full-thickness incisional surgical treatments used for angle-closure glaucoma are the same as those used for open-angle glaucoma. However, laser trabeculoplasty is relatively contraindicated if the angle is so narrow that additional peripheral anterior synechiae may form after the laser procedure. Typically, partial-thickness procedures are not indicated.

Prevention of Angle-Closure Glaucoma

Acute angle-closure glaucoma can be prevented by avoiding pupillary dilation in people at risk. This is important when considering applying topical agents to dilate the eye for examination (eg, [cyclopentolate](#), [phenylephrine](#)) or for treatment (eg, homatropine) or when giving systemic medications that have the potential to dilate the pupils (eg, [scopolamine](#), alpha-adrenergic agonists commonly used to treat urinary incontinence, medications with anticholinergic effects).

Key Points

- Angle-closure glaucoma can develop acutely, intermittently, or chronically.
- Suspect acute angle-closure glaucoma based on clinical findings and confirm it by measuring intraocular pressure.
- Confirm chronic angle-closure glaucoma by documenting peripheral anterior synechiae and optic nerve and visual field changes.
- Treat acute angle-closure glaucoma as an emergency.
- Consult an ophthalmologist to arrange laser peripheral iridotomy for all patients with angle-closure glaucoma.

Drug Information for the Topic



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